INCH-POUND

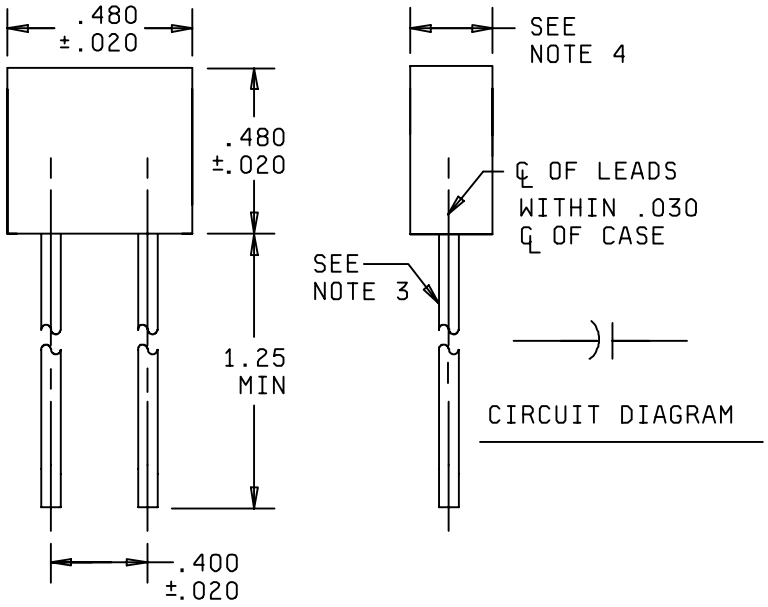
MIL-PRF-20/37E
21 June 2001
SUPERSEDING
MIL-PRF-20/37D
30 June 1988

PERFORMANCE SPECIFICATION SHEET

# CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR07 AND CC07

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-PRF-20.




| Inches | mm |
|---|---|
| .020 | 0.51 |
| .030 | 0.76 |
| .400 | 10.16 |
| .480 | 12.19 |
| 1.250 | 31.80 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Lead diameter shall be .025 +.004, -.001 (0.64 +0.10, -0.03 mm).
4. Thickness shall be .200 max (5.08 mm) for 100,000 pF capacitors or .140 ± .010 (3.56 ± 0.25 mm) for all other capacitors.

FIGURE 1. Styles CCR07 and CC07 capacitors.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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Lead type: Radial.

Case type: Molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Failure rate level (CCR07 only): M (1.0 percent), P (0.1 percent), R (.01 percent) or S (.001 percent).

Thermal shock and voltage conditioning (CCR07 only): In accordance with MIL-PRF-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-PRF-20.

Dielectric withstanding voltage: In accordance with MIL-PRF-20.

Body insulation: Test II.

TABLE I. Capacitor characteristics.

| Type designation 1/ 2/ | Rated voltage (volts, dc) | Nominal capacitance (pF) | Capacitance tolerance | Type designation 1/ 2/ | Rated voltage (volts, dc) | Nominal capacitance (pF) | Capacitance tolerance |
|---|---|---|---|---|---|---|---|
| CC-07CG222-- | 200 | 2,200 | FGJK | CC-07CG183-- | 50 | 18,000 | FGJK |
| CC-07CG272-- | 200 | 2,700 | FGJK | CC-07CG223-- | 50 | 22,000 | FGJK |
| CC-07CG332-- | 200 | 3,300 | FGJK | CC-07CG273-- | 50 | 27,000 | FGJK |
| CC-07CG392-- | 200 | 3,900 | FGJK | CC-07CG333-- | 50 | 33,000 | FGJK |
| CC-07CG472-- | 200 | 4,700 | FGJK | CC-07CG393-- | 50 | 39,000 | FGJK |
| CC-07CG562-- | 100 | 5,600 | FGJK | CC-07CG473-- | 50 | 47,000 | FGJK |
| CC-07CG682-- | 100 | 6,800 | FGJK | CC-07CG563-- | 50 | 56,000 | FGJK |
| CC-07CG822-- | 100 | 8,200 | FGJK | CC-07CG683-- | 50 | 68,000 | FGJK |
| CC-07CG103-- | 100 | 10,000 | FGJK | CC-07CG823-- | 50 | 82,000 | FGJK |
| CC-07CG123-- | 100 | 12,000 | FGJK | CC-07CG104-- | 50 | 100,000 | FGJK |
| CC-07CG153-- | 50 | 15,000 | FGJK | | | | |

1/ Complete type designation will include the following:
1st dash - Symbol “R” (for style CCR07) or dash will be deleted (for style CC07).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR07 only) or dash will be deleted (for style CC07).

2/ Capacitors deleted since previous revisions shall be replaced by selecting a capacitor with the nearest parameter.



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Solderability: In accordance with MIL-PRF-20.

Resistance to soldering heat: In accordance with MIL-PRF-20.

Life: In accordance with MIL-PRF-20, operating condition 2.

Marking: In accordance with MIL-PRF-20, method I.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AV
Navy - AS, MC, OS, SH

Preparing activity:
DLA - CC

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